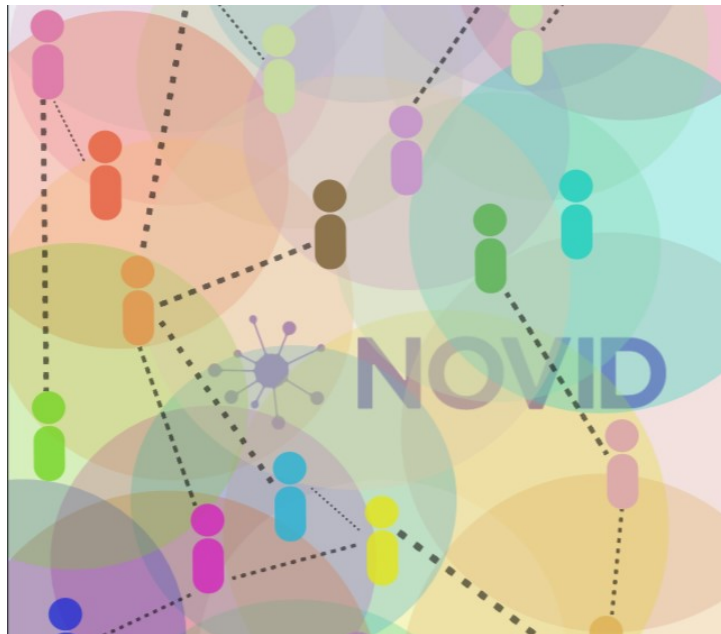


NOVID: Exploring the Network Theory Used to Track the Spread of COVID-19

Richard Wang



1 Introduction

NOVID was the first contact tracing app in the world to anonymously trace users' exposure to COVID-19. But how did the Carnegie Mellon University team led by professor Po-Shen Loh develop NOVID using network theory?

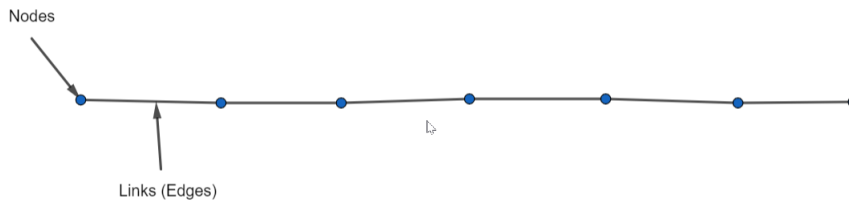
2 Network Theory

Network theory is a technique used to analyze graphs by turning them into complex systems. A network is a graph with two components, the node and the edge. A node (vertex) is a common point connected by branches. In this case, nodes are people. A branch is a line segment that connects two nodes. Edges (links) are lines joining a pair of nodes. Degrees are the number of edges that connect to a vertex. There are multiple kinds of graphs, but only simple graphs are used in this article.

We can analyze the math behind the spread of COVID-19 by considering three different situations.

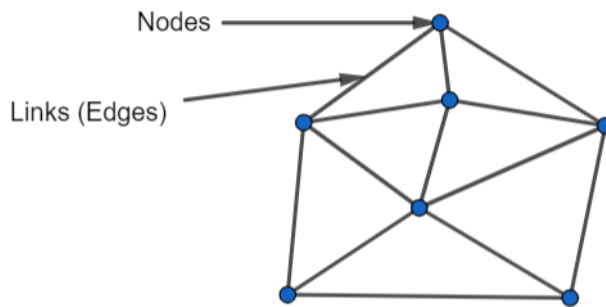
2.1 Situation 1: Weird Society

Imagine a society where the people are organized in a straight line and only contaminate the people next to them—two people in the middle or one person on the edge. The spread would be linear because an infected person can only contact one person on either side at a time.



2.2 Situation 2: Normal Society

However, in a normal society where people spread diseases to others closest to them, growth is exponential. Given a random set of points linked with each other, COVID-19 spreads to the points closest. It takes the second closest point a degree of two to spread COVID-19 to them. It would take the third closest point a degree of three to spread COVID-19 to them, etc. However, the graph also depends on how long people spend with others. In order to stop the spread, links must be closed.



2.3 Situation 3: Accurate Society

What if the graph were not only based on how close people are to others, but also on how long people are in contact with others? That would be a much more accurate analysis of the network for how COVID-19 spreads. Therefore, links that aren't limited by the nodes they are next to can give even more information.

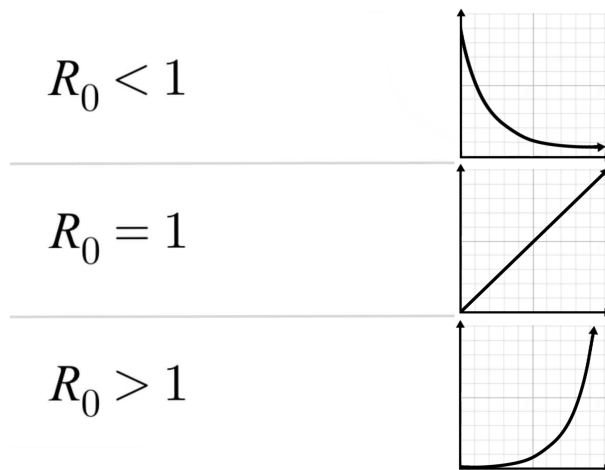
Let's say that you are notified whether there are any infected points nearby you or if you are infected. The more time people spend with others, the more the COVID-19 infection spreads and

the probability of you getting infected increases.

3 Graphing

R_0 , or R naught, which represents the reproduction number, shows the growth, linear, or decay pattern of a graph. It can be simplified into the following information:

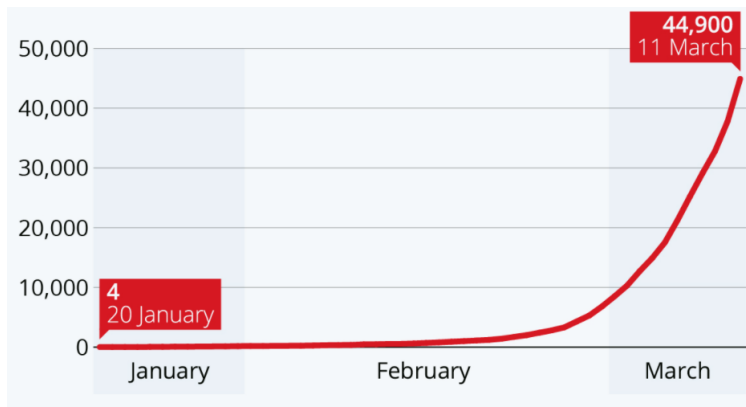
If R_0 is less than 1, the graph shows decay. If R_0 is equal to 1, the graph is linear. If R_0 is greater than 1, the graph is exponential.



In terms of disease transmission, R_0 indicates how contagious the disease is. It quantifies the number of people to which an infected person can spread the disease over time.

If R_0 equals 1, each existing infection causes one new infection. The disease will stay alive and stable, but there won't be an outbreak or an epidemic. These circumstances were described in Situation 1. If R_0 is greater than 1, each existing infection causes more than one new infection. The disease is transmitted between people, and there may be an outbreak or epidemic. This is the case outlined in Situation 2.

The R_0 for COVID-19 in April 2020 was 5.7, according to a study published online on *Emerging Infectious Diseases*, an open-access, peer reviewed journal of the Center for Disease Control and Prevention (CDC). That's about double an earlier R_0 estimate of 2.2 to 2.7. The 5.7 means that one person with COVID-19 can potentially transmit the coronavirus to 5 to 6 people, rather than the 2 to 3 researchers originally hypothesized. Researchers calculated the new number based on data from the original outbreak in Wuhan, China. With an R_0 of 5.7, at least 82 percent of the population needs to be immune to COVID-19 to stop its transmission through vaccination and herd immunity.



The image above shows the actual recorded data of how COVID-19 spread throughout China in early 2020.

How can we use this to help us stop the spread of COVID-19? In terms of network theory, we want R_0 to be less than 1, so that each existing infection causes less than one new infection. Therefore, the infection would eventually decline and die out.

4 Contact Tracing

Contact tracing quantifies the following factors to decrease the infection rate:

4.1 Infectious Period

The infectious period is the amount of time a person has been in contact with other people. The longer the infectious period, the greater probability of infection, and the higher the R_0 .

4.2 Mode of Transmission

The mode of transmission is how the disease is spread. Because COVID-19 is an airborne infection, it spreads more quickly than other diseases that depend on physical contact to be transmitted. Therefore, COVID-19 has higher R_0 values than many other types of infections.

4.3 Contact Rate

The contact rate is how many people an infected person has been in contact with over time. This variable is not specific compared to the last two factors because it depends on multiple factors like location, public health measures, quarantines, and travel bans.

In short, contact tracing involves asking a person whether they're infected and if they've been with others who are infected. If either choices are true, quarantine them. From the principle idea of contact tracing, it is helpful for people to be notified whether they are possibly infected. It is to protect other people from you, and to protect you from others.

Contact tracing relies on network theory. If any node is close to another node, the first node can be notified to avoid the others in hopes of safety. This cancels any connections (edges) and isolates every node from risk of infection.

The risk of spreading COVID-19 can be reduced by notifying people to distance themselves from those who are infected and to distance from others if they are infected themselves.

5 Bluetooth-based apps

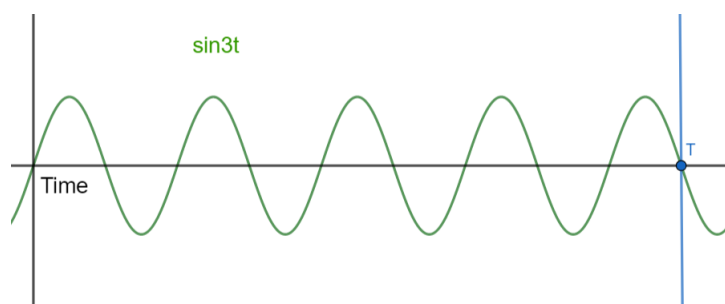
How can we use the information gained through contact tracing and network theory for practical use? We could hire volunteers to record whom infected people have contacted, but it is very inconvenient to record every single infected person. We can allow people to record and survey daily who they've contacted, but then again, it won't be convenient.

The best and most effective way to make contact tracing possible for everyone is by creating a mobile app. The contact tracing app NOVID uses sound waves and Bluetooth to calculate how far users of the app are from other users. This is extremely helpful in fighting off COVID-19, because there is no other contact tracing app in the world that directly tells you how far you are from others.

How does the app work? Given two phones, both phones open their mics. One sends a sound signal to the other and the second phone closes their mic and analyzes how long it took to receive the sound. The app notifies whether or not someone is at a maximum of 9 feet away. The 9 feet takes into account whether or not you're holding your phone, which decreases the distance between two phones due to arm length.

5.1 Mathematics of the Sound Signal

Although the app can potentially prevent COVID-19 infections, there are still imperfections to the app. If, for example, you are in your house, the sound can bounce off the walls and need the structure of the room in order to calculate how the sound signals are traveling. In addition, the phone must pick the proper frequency of the sound signal. Unfortunately, the mathematics in how sound signals are calculated despite being bounced off of surfaces is not publicly displayed. However, the mechanics for finding the proper frequency can still be calculated.



The picture above gives an example of estimates for what a phone would pick up if it is expecting a signal. How can we tell that this signal is the same signal from the NOVID app? In other words, how can we tell that this is the proper frequency the phone catches?

In the graph, the function of the graph is $\sin 3T$. If we calculate the integral (area under the curve) of the graph from 0 to T, then the area and T have a linear relationship. What if we calculate the integral of $f(x)\sin 2T$ from 0 to T? You can calculate it by substituting as follows:

$\sin 3t * \sin 2t = 1/2[\cos(3t - 2t) - \cos(3t + 2t)] = 1/2[-\cos 5t]$ which becomes

$$1/2 * \int_0^T \cos(t) dt - 1/2 \int_0^T \cos(5t) dx.$$

This value approaches a constant. Therefore, we can derive any sound signal with the same frequency by calculating whether or not the function of the signal has the same value as the constant.

6 Conclusion

In conclusion, with the help of mathematics, the NOVID app was developed as an effort to limit COVID-19 infections. From network theory to analyzing different social models to graphing the spread of COVID-19 to finding the factors of infection then to how to stop the spread by using an app, a web of relations emerged from the mathematics we study at school.

References

- Achaiah, Nithya C, Sindhu B Subbarajasetty, and Rajesh M Shetty. "R0 and Re of COVID-19: Can We Predict When the Pandemic Outbreak Will Be Contained?" Indian Journal of Critical Care Medicine. Jaypee Brothers Medical Publishers, November 2020. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7751056/>.
- Alguliyev, Rasim, Ramiz Aliguliyev, and Farhad Yusifov. "Graph Modelling for Tracking the COVID-19 Pandemic Spread." Infectious Disease Modelling.
- "ATrain Education." 3. The Basic Reproduction Number: R Naught — ATrain Education. Accessed July 2, 2021. <https://www.atrainceu.com/content/3-basic-reproduction-number-r-naught>.
- Block, Per, Marion Hoffman, Isabel J. Raabe, Jennifer Beam Dowd, Charles Rahal, Ridhi Kashyap, and Melinda C. Mills. "Social Network-Based Distancing Strategies to Flatten the COVID-19 Curve in a Post-Lockdown World." Nature News. Nature Publishing Group, June 4, 2020. <https://www.nature.com/articles/s41562-020-0898-6>.
- Elsevier, December 13, 2020. <https://www.sciencedirect.com/science/article/pii/S2468042720301068>.
- "Network Theory." Network Theory - an overview — ScienceDirect Topics. Accessed July 2, 2021. <https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/network-theory>.
- Ramirez, Vanessa Bates. "What Is R0? Gauging Contagious Infections." Healthline. Healthline Media, April 20, 2020. <https://www.healthline.com/health/r-nought-reproduction-number>.
- "Understanding Predictions: What Is R-Naught?" Harvard Global Health Institute, July 8, 2020. <https://globalhealth.harvard.edu/understanding-predictions-what-is-r-naught/>.